

Topic 18B: Amines, amides, amino acids and proteins

8. be able to identify:
 - i the amine and amide functional groups
 - ii molecules that are amino acids
9. understand the reactions of primary aliphatic amines, using butylamine as an example, with:
 - i water to form an alkaline solution
 - ii acids to form salts
 - iii ethanoyl chloride
 - iv halogenoalkanes
 - v copper(II) ions to form complex ions
10. understand reasons for the difference in basicity of ammonia, primary aliphatic and primary aromatic amines given suitable data
11. understand, in terms of reagents and general reaction conditions, the preparation of primary aliphatic amines:
 - i from halogenoalkanes
 - ii by the reduction of nitriles
12. know that aromatic nitro-compounds can be reduced, using tin and concentrated hydrochloric acid, to form amines
13. understand that amides can be prepared from acyl chlorides
14. know that the formation of a polyamide is a condensation polymerisation reaction
15. be able to draw the structural formulae of the repeat units of condensation polymers formed by reactions between:
 - i dicarboxylic acids and diols
 - ii dicarboxylic acids and diamines
 - iii amino acids
16. understand the properties of 2-amino acids, including:
 - i acidity and basicity in solution, as a result of the formation of zwitterions
 - ii effect of aqueous solutions on plane-polarised monochromatic light
17. understand that the peptide bond in proteins:
 - i is formed when amino acids combine, by condensation polymerisation
 - ii can be hydrolysed to form the constituent amino acids, which can be separated by chromatography

CORE PRACTICAL 15: Analysis of some inorganic and organic unknowns